

Shuvam Banerji Seal

Computational Chemist | AI/ML & Information Retrieval Researcher | DeepTech & AI-Fintech Co-Founder

BS-MS Student (Chemistry Major, Computer Science Minor)

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Research Interests

Computational Chemistry and Materials Modelling (density functional theory, Quantum ESPRESSO, Gaussian, molecular dynamics with LAMMPS) · Machine Learning for the Physical and Life Sciences (transformers, mixture-of-experts, surrogate models for simulation) · Structural Bioinformatics and Genomic Data Analysis (RNA and protein structure prediction, sequence-space encodings) · Information Retrieval and Retrieval-Augmented Generation (BM25, dense retrieval, hybrid search, domain-specific RAG) · Formal Methods for Scientific Reasoning (Lean 4, machine-checked proofs) · AI Safety, Evaluation and Agentic Systems (sycophancy detection, LLM-as-a-Judge, interpretability) · Information Theory.

Publications

AgriIR: A Scalable Framework for Domain-Specific Knowledge Retrieval

ECIR 2026

Shuvam Banerji Seal, Aheli Poddar, Alok Mishra, and Dr. Dwaipayan Roy

- Developed a scalable information retrieval framework specifically designed for agricultural domain knowledge accessibility in developing regions, implementing advanced retrieval algorithms optimized for domain-specific terminology and contextual understanding to advance retrieval systems for social good.
- DOI: 10.1007/978-3-032-21324-2_37

Hierarchical Opinion Classification using Large Language Models

FIRE 2025

Shuvam Banerji Seal, Alok Mishra, and Utkarsha Ghosh

- Developed a novel parameter-efficient fine-tuning approach for the Gemma LLM using a custom two-layer classification head and class-weighted cross-entropy loss to address data imbalance. Reformulated three-level hierarchical opinion labels into an 8-class flat scheme, exploring dual methodologies like selective fine-tuning and instruction-tuning to advance LLM adaptation for hierarchical classification under computational constraints.
- <https://ceur-ws.org/Vol-4173/T10-3.pdf>

IISERK@ ToT_2024: Query Reformulation and Layered Retrieval for Tip-of-Tongue Items

TREC 2024 Proceedings

Shuvam Banerji Seal & Subinay Adhikary, Soumyadeep Sar, and Dr. Dwaipayan Roy

- Explored various approaches for known-item retrieval in the TREC 2024 Tip-of-the-Tongue track, focusing on retrieving previously encountered items when searchers struggle to recall exact identifiers
- Implemented four-step query reformulation technique combined with two-layer retrieval using BM25 with optimized parameters and Large Language Models for query enhancement Achieved best performance with two-layer retrieval approach, obtaining Recall@1000 of 0.8067 and enhanced NDCG metrics through systematic query reformulation
- <https://trec.nist.gov/pubs/trec33/papers/IISER-K.tot.pdf>

SycoLex: A Cross-Jurisdictional Benchmark for Detecting Sycophancy in Legal Case Reasoning

CIKM 2026 Resource Track

(under review)

Shuvam Banerji Seal and collaborators, IISER Kolkata

- Constructed a 1,954-case, three-jurisdiction benchmark (US Supreme Court, Indian Supreme Court, Indian Consumer Courts) with six sycophancy-inducing prompt variants, responses from five LLMs, and paired LLM-as-Judge plus human-expert annotations.
- Released openly on Hugging Face under CC-BY-4.0; forms the evaluation resource for the FIRE 2026 SYCOPHANCY shared task.
- huggingface.co/datasets/ShuvBan/SycoLex

Computational Modeling of [VO(SALIEP)(DTP)] as Water Reducing Catalyst

2024-25 (to be submitted)

Shuvam Banerji Seal, Guided by Dr. Soumyajit Roy

- Implemented DFT methods (B3LYP) for MO energy calculations and electronic structure analysis using Gaussian and conducted mechanistic studies using transition state theory and reaction pathway analysis catalyst performance prediction for water reduction

Master's Thesis (Ongoing)

Project I – Gray-Code / Karnaugh-Map Structure of Nucleic-Acid and Protein Sequences, Formalised in Lean 4 2026 – Present
BS-MS Thesis, IISER Kolkata

- Encoding DNA, RNA, and protein sequences into binary sequence space and organising the resulting codewords on **Gray-code** orderings and **Karnaugh maps**, so that adjacency in the map corresponds to a single-symbol mutation in the biological sequence.
- Extracting the binary relations, symmetries, and distance structure induced by this encoding, and testing whether they predict biologically meaningful structure (codon degeneracy, mutation neighbourhoods, amino-acid physicochemical classes).

- Formalising and machine-checking every theorem in the **Lean 4** proof assistant (Mathlib), giving fully rigorous proofs of the structure-based correspondence between the pure combinatorial formulation and real genomic sequences.
- **Keywords:** formal verification, Lean 4, theorem proving, Gray code, Boolean algebra, combinatorics on words, information theory, genomics, bioinformatics.

Project II – A Transformer Mixture-of-Experts Model for RNA / Nucleic-Acid Structure Prediction

2026 – Present

BS-MS Thesis, IISER Kolkata

- Building a **Mixture-of-Experts (MoE) transformer** for nucleic-acid (RNA) secondary and tertiary structure prediction, with sparse expert routing so that specialised experts handle distinct structural motifs (helices, loops, pseudoknots) at a fixed inference budget.
- Designing the sequence-to-structure pipeline end to end: tokenisation of nucleotide sequences, evolutionary/MSA features, geometry-aware attention, distance-map and torsion-angle prediction heads, and structure-quality losses.
- Training and evaluation on public RNA structure benchmarks with multi-GPU distributed training, mixed precision, and systematic ablation of routing strategies and expert counts.
- **Keywords:** deep learning, transformers, mixture-of-experts, RNA structure prediction, structural bioinformatics, PyTorch, distributed training.

Research Experience

A Hybrid RAG Architecture for Verifiable Answer Synthesis with Automated Source Citation

July 2025

Under Dr. Dwaipayan Roy, Dept. of Computation and Data-Science, IISER-Kolkata

- Developed a comprehensive RAG pipeline for IISER-K intranet documents, including automated data acquisition via Selenium, OCR processing with PyMuPDF and EasyOCR, and text extraction. Implemented context-aware text chunking with metadata enrichment and high-dimensional embeddings using Qwen3-4B model, stored in FAISS vector database. Designed hybrid search strategy combining dense vector retrieval with BM25 keyword search, fused using Reciprocal Rank Fusion for optimal relevance. Integrated query refinement with hypothetical answers and verifiable inline citations in a Streamlit application for transparent, grounded responses.

Developing an Advanced Retrieval Model for TREC-Tip of Tongue Queries

2024 (TREC 2024 Proceedings)

Guided by Dr. Dwaipayan Roy, Dept. of Computation and Data-Science, IISER-Kolkata

- Engineered multi-layer BM-25 filtering system (in Lucene, Java) with dynamic search-domain contraction based on extracted keywords and implemented transformer-based query expansion and semantic matching using Local LLMs through multi-shot and chain-of-thought prompting achieving results on par with specifically-modified DPR models.

Industrial Experience

Research Intern, HistoXai (Astroloop Technologies Pvt. Ltd.)

May 2025 - July 2025

Bangalore

- Conducted an extensive comparative analysis of digital histopathology slide quality assessment tools, reviewing 30+ open-source frameworks (e.g., HistoQC, PathProfiler, GrandQC, HistoROI, FASTPathology) to evaluate architectural designs, datasets, performance metrics, and limitations for future model integration.
- **Technologies:** Python, OpenCV, PyTorch, Scikit-learn, Pandas, Literature Survey Tools

Open Datasets Released (Hugging Face)

Hugging Face Profile

huggingface.co/ShuvBan

Public research datasets (further datasets currently private / pre-release)

- **SycLex** – Benchmark for detecting sycophancy in legal case reasoning. 1,954 real legal cases across three jurisdictions (US Supreme Court 300, Indian Supreme Court 1,500, Indian Consumer Courts 154), six sycophancy-inducing prompt variants, responses from five LLMs, LLM-as-Judge and human expert annotations. Tasks: text classification, text generation. License CC-BY-4.0.
- **AgriIR_dataset** – 15,247 curated Indian agricultural knowledge entries with quality scores and domain metadata (crops, soils, regions, farming practices), released alongside the AgriIR paper (ECIR 2026; [arXiv:2604.16353](https://arxiv.org/abs/2604.16353)). License CC-BY-4.0.
- Datasets are versioned, documented with dataset cards, and used as the public evaluation resources for the SYCO PHANCY (FIRE 2026) shared task and the AgriIR retrieval framework.

Entrepreneurial Ventures

Co-Founder & Lead Developer, iFiNN

2025 – Present

Incubated at RISE Foundation, IISER Kolkata

- Co-founded an AI-driven fintech startup funded by **MeitY Startup Hub (MSH)** through the **GENESIS** program.
- Developing a user-friendly platform with simple UI/UX to democratize access to essential ML and AI tools for automated smart trading alerts across stocks, crypto, and mutual funds.
- **Status:** Funded; DPIIT registration in progress (Q1 2026); Product shipping by 2027.

Co-Founder & CTO, UnderWater AI

2025 – Present

Incubated at RISE Foundation, IISER Kolkata

- Co-founded a deeptech startup funded by **MeitY Startup Hub (MSH)** through the **GENESIS** program.
- Developing software solutions to enhance underwater image quality using deep neural networks and early fusion learning for marine species identification, serving researchers and industrial marine applications.

- **Status:** Funded; DPIIT registration in progress (Q1 2026); Product shipping by 2027.

Flagship Open-Source Software & Research Frameworks

Molecule3D / LAMMPS Web GUI – Browser-Native Molecular Simulation Workbench

2026 – Present

Minimum viable product; free, open-source (MIT), GPU-accelerated, runs entirely client-side

- Built a complete **LAMMPS** workbench in the browser that lowers the entry barrier to molecular dynamics: a visual **Script Builder** (drag-and-drop flowchart over a 186-command library covering the full general-command surface, locked by an automated coverage test against docs.lammps.org), a **Compiler Helper** that turns 27 CMake build options (Kokkos, GPU back end/precision/arch, FFT/heFFTe, OpenMP, FFMPEG) into ready-to-run LAMMPS build scripts, and a 3D **Structure Viewer**.
- Reads LAMMPS data, XYZ, PDB and CIF structures plus LAMMPS dump trajectories; supports trajectory playback, in-browser MP4/WebM recording, distance/angle/dihedral measurement, and an **Analysis** module computing PBC-aware RDF $g(r)$, MSD, density profiles and speed histograms with CSV export.
- Round-trips existing in.* input scripts back into an editable flowchart, exports presentation-quality SVG/PNG pipeline diagrams, and persists all state locally – nothing is uploaded.
- **Stack:** TypeScript, React, Three.js / React Three Fiber, WebGL, Vite, Vitest (143 passing tests), GitHub Actions CI/CD.

AgriIR – Configurable RAG Framework for Domain-Specific Retrieval

2025 – 2026

Reference implementation of the ECIR 2026 paper

- Six-stage configurable retrieval-augmented generation pipeline with deterministic citations and domain-specific agents, built on FAISS, Ollama-served open-weight LLMs, and hybrid BM25 + dense retrieval.
- Ships the accompanying 15k-entry open dataset and a reproducible evaluation harness. DOI: [10.1007/978-3-032-21324-2_37](https://doi.org/10.1007/978-3-032-21324-2_37)

IISERKonnnect – Campus Super-App for IISER Kolkata

2026 – Present

Sole developer; native Android (Kotlin) + web client. In campus-wide testing with 2,000+ users at IISER Kolkata

- Designed and built a single application that unifies the entire IISER Kolkata intranet surface – WeLearn courses, files, assignment deadlines and attendance; mess transactions, budget analytics and daily menu; academic calendar and campus events; grade cards; the previous-year-question archive; student and administrative notice boards; the library catalogue with live availability; the ePrints research archive; the VoIP directory; the TCP counter and MAC registration; and a network health dashboard.
- Implemented every campus service as a hand-written HTML parser (Kotlin/Jsoup on Android, ported 1:1 to DOMParser on web), with captcha-backed login flows, per-session cookie jars, and a strict campus-only host allow-list; credentials never leave the device.
- Added a serverless **Campus Chat** over WebRTC data channels (DTLS end-to-end encrypted, invite-code handshake, optional STUN for strict NATs), replacing the earlier raw-TCP discovery layer.
- **Stack:** Kotlin, Jetpack Compose, Jsoup, OkHttp, JavaScript (ES modules), Node.js companion proxy, WebRTC, Playwright end-to-end tests.

Introduction to Molecular Simulation – Open Talk & Hands-On Companion

2026

Most-starred public repository; Marp slide deck, concept notes and runnable examples

- Authored a complete beginner-to-running-simulation course: statistical-mechanics motivation, force fields, MD integrators/thermostats/barostats vs. Monte Carlo, NVE/NVT/NPT ensembles, and observable extraction (RDF, MSD, VACF, free energy), with reproducible LAMMPS simulations used as the demo trajectories for Molecule3D.

DFT Notes – Open Knowledge Base on Density Functional Theory

2026

Open educational resource

- Progressive chapters from the Schrödinger and many-body problem through Hartree-Fock, the Hohenberg-Kohn theorems, the Kohn-Sham equations and exchange-correlation functionals, each with runnable Python and generated plots, plus a Mermaid-driven knowledge graph of the chapter dependencies.

Fernholz Stochastic Portfolio Theory Python Library

Ongoing

Quantitative Finance Library

- Developing a Python library for Stochastic Portfolio Theory, implementing diversity-weighted portfolios and relative arbitrage strategies using stochastic calculus and continuous semimartingales.

Tutorials Developed

Complete C Programming Course

2025

Open Source Curriculum

- Created a comprehensive 20-module C programming curriculum covering fundamentals to advanced topics including Network Programming, Machine Learning, and GUI development with GTK4.

Python Course for Beginners

2025

Open Source Curriculum

- Designed a progressive Python learning resource spanning core concepts to local LLM deployment, featuring interactive notebooks, real-world project implementations, and database integration.

Shaders for Beginners

2026

Open Source Curriculum with interactive WebGL editor

- Wrote 11 progressive lessons teaching GPU shader programming from first principles – OpenGL and GLSL, the graphics

pipeline, an introduction to Vulkan, and audio-reactive shaders – with 15+ worked shaders and C/C++ host implementations.

CH4114 Molecular Simulation – Hands-On Tutorials

2026

Course tutorial material, IISER Kolkata

- Prepared and delivered tutorial sessions on reproducible computational practice for molecular simulation: Git and GitHub, project organisation, and uv-based Python environment management.

Projects

Solvation of C₆₀ Nanoparticles in Water

2025

Computational Science Project

- Ran LAMMPS molecular dynamics of C₆₀ solvation with AIREBO potentials, automated structural feature extraction (RDF, coordination numbers), and trained a surrogate ML pipeline (Gaussian process regression, neural networks, XGBoost) to predict system properties at unsampled parameter values.

opngx – High-Speed Camera Footage Extractor

2026

Scientific Instrumentation Tooling

- Wrote a pixel-exact extractor for Optronis .bin high-speed-camera recordings that saturates all CPU cores with runtime SIMD dispatch, shipped as a CLI, a Tk GUI and a Python library for Linux and Windows.

TCP Activity Monitor for Arch Linux

2025

System Utility

- Developed a system-wide TCP monitoring utility for Arch Linux integrating systemd services to log kernel statistics, socket states, and per-process connections, enhancing network diagnostic capabilities.

WeLearn Bot in C

2025

Utility Application

- Engineered a multi-threaded educational resource automation tool in C featuring both CLI and GTK4 GUI interfaces with secure session management and encrypted credential storage for efficient bulk downloads.

IndicAgri: AI Platform for Indian Agriculture

Aug 2025

Capital One Launchpad Hackathon 2025

- Architected a multi-modal RAG chatbot for Indian agriculture supporting 20+ languages using Gemma/DeepSeek LLMs and Indic-Conformer, backed by an autonomous 15k+ document curation pipeline.

Automated Event Coupon Management System

2025

Event Automation System

- Built a secure Flask-based event management system featuring Google OAuth 2.0 authentication, automated bulk emailing, and real-time QR code verification with mobile scanner integration.

Legal Document Retrieval RAG Application

Feb 2025

Legal Tech RAG System

- Developed a legal document retrieval RAG stack leveraging LLaMA-3.2, Nomic embeddings, and ChromaDB, with a Streamlit interface and Mistral-OCR for processing complex legal texts.

Full-Stack E-Commerce GUI in C

2024

Under Dr. Kripabandhu Ghosh, IISER-Kolkata

- Implemented a full-stack E-Commerce GUI in C using GTK4 and SQLite3, incorporating a static context chatbot with optimized BM25 retrieval for intelligent customer interaction.

Wi-Fi Channel Optimizer

2024

Network Optimization Tool

- Engineered a Bash-based Wi-Fi optimization script to automate network scanning, speed benchmarking, and channel selection, significantly enhancing wireless network performance on Linux systems.

SMC Canteen System Modernization

2024

IISER-Kolkata

- Modernized the IISER-Kolkata Canteen System by porting legacy code to a modular Django-based architecture, improving maintainability and scalability for 2000+ users.

Coordination Chemistry Twists Simulator

Nov 2024

Computational Chemistry Simulation

- Developed a Python molecular dynamics simulator for visualizing Ray-Dutt and Bailar twist mechanisms in coordination complexes, providing interactive 3D stereochemical analysis.

Agentic Database Builder

Aug 2024

Autonomous Data Engineering

- Designed an AI-driven agentic system for autonomous database construction, utilizing LLM decision-making for automated schema generation, data validation, and scalable curation.

Interactive Spherical Harmonics Visualizer

2023-24

Scientific Visualization Engine

- Built an interactive 3D quantum orbital visualizer using Flask and Plotly, enabling real-time computation and rendering of spherical harmonics for educational visualization.

More projects can be found at my [Portfolio Website](#) or [Github Profile](#).

Technical Skills

Core Competencies : Technical Leadership | Research & Development | Algorithm Development | Scientific Software Engineering | Strategic Planning | Event Management | Mentoring | Public Speaking | Cross-functional Collaboration

Research Computing : Information Retrieval (Apache Lucene, BM25, Reciprocal Rank Fusion) | High-Performance Computing (HPC) | Parallel Computing (OpenMP, MPI, CUDA) | Numerical Methods | Statistical Mechanics | Information Theory | Reproducible Research

Computational Chemistry & Materials : Density Functional Theory (DFT) | **Quantum ESPRESSO**

(plane-wave/pseudopotential SCF, geometry relaxation, band structure, DOS, phonons) | Gaussian (B3LYP, basis sets, transition-state search) | VASP workflows | ORCA | Molecular Dynamics (LAMMPS, GROMACS, VMD, OVITO) | Force Fields (AIREBO, ReaxFF, EAM, Lennard-Jones) | Monte Carlo | Ab-Initio Electronic Structure | Exchange-Correlation Functionals | Radial Distribution Functions, MSD, Free-Energy Analysis

Structural Biology & Bio-Informatics : Protein and Nucleic-Acid Structure Prediction | RNA Secondary/Tertiary Structure | Sequence Analysis (DNA, RNA, Protein) | PyMol | ChimeraX | AutoDock Vina | PyDock | Molecular Docking | Genomic Data

Analysis

Machine Learning & Deep Learning : PyTorch | TensorFlow | Transformers | Mixture-of-Experts (MoE) | Attention Mechanisms | Parameter-Efficient Fine-Tuning (LoRA/QLoRA) | Distributed & Mixed-Precision Training | Gaussian Process Regression | XGBoost | Scikit-learn | Model Evaluation & Ablation Studies

LLMs, RAG & Agentic AI : Retrieval-Augmented Generation (RAG) | Dense & Sparse Hybrid Retrieval | Vector Databases (FAISS, ChromaDB) | LangChain | Ollama | vLLM | HuggingFace Hub & Datasets | Prompt Engineering (multi-shot, chain-of-thought) | Autonomous Agents | LLM-as-a-Judge | Sycophancy & Bias Evaluation | AI Safety | Qwen2/3 | Gemma (1B, 27B) | DeepSeek | LLaMA-3.2 | Nomic Embeddings | Mistral-OCR | Indic-Conformer

Formal Methods : Lean 4 & Mathlib | Machine-Checked Theorem Proving | Formal Verification | Boolean Algebra & Gray Codes | Combinatorics on Words

Programming Languages : Python | C/C++ | Kotlin | TypeScript/JavaScript | Java | Rust | Fortran | Lean 4 | Bash | SQL | QBASIC | GWBASIC

Python Libraries - Core & Scientific : NumPy | Pandas | SciPy | Matplotlib | Plotly | Scikit-learn | OpenCV | PyTorch | TensorFlow | ASE | MDAnalysis

Python Libraries - NLP, Data & Documents : HuggingFace Transformers & Datasets | LangChain | NLTK | SpaCy | FAISS | ChromaDB | Selenium | BeautifulSoup | PyMuPDF | EasyOCR | Streamlit | Manim

Digital Pathology Tools : HistoQC | PathProfiler | GrandQC | HistoROI | FASTPathology

Development Stack : Full Stack (HTML/CSS/JavaScript) | React | Three.js / React Three Fiber | WebGL & GLSL | Vite | Django | Flask | Node.js | REST APIs | WebRTC | Databases (PostgreSQL, MongoDB, MySQL, SQLite3) | Git & GitHub Actions (CI/CD) | Docker | CMake | Make | Vitest | Playwright

Mobile Development : Android (Kotlin) | Jetpack Compose | Coroutines & Flow | Room | OkHttp | Jsoup

GUI Development : GTK4 in C | Qt in C++ | Glade | Tkinter

Operating Systems & Infrastructure : Linux (Arch, Ubuntu) | Shell & systemd | SSH, OpenSSL | Server & Cluster Handling | SLURM-style Job Submission | GPU Computing (CUDA)

Laboratory Instrumentation : UV-Vis | ATR-FTIR | TGA | DSC | Optical Bench | XRD | GC-MS | Column Chromatography | Fluorimeter | SEM | TEM | AFM

Additional Tools : $\LaTeX$ & Beamer | Marp | UI/UX Design (Figma) | Technical Writing | Scientific Visualisation | HTML E-mailing

Achievements & Honors

Hackathons:

Capital One Launchpad (Top 14 Team among 5000+ Teams) 

2025

Developed IndicAgri: Multi-modal RAG Platform for Indian Agriculture (Team Fibonacci)

Capital One India

- Selected as one of the Top 14 teams from 5,073 registered teams across India in Capital One's flagship AI/ML innovation challenge
- Built an agentic AI-powered agricultural advisor supporting 20+ Indian languages with scientifically cited, hyper-localized outputs using RAG, LangChain, and open-source LLMs (Gemma, DeepSeek)
- Engineered autonomous data pipeline creating a novel 15k+ document dataset for Indian agriculture, released on Hugging Face

StatusCode1 (Awarded 1st Rank in GIAN Track) 

2024

Developed an AI-based Search Engine for GIAN's Database For Abandoned US Patents

IIIT-Kalyani

- A searching algorithm based on Nomic Embeddings of the patent abstracts and similarity computations thus enabling the user to search for patents in natural language without the use of specific terminologies.
- Also created a web-scraping algorithm to get the patent data using Selenium, BeautifulSoup and fake_useragent for anti-scraping measures by the website

StatusCode0 (Awarded 1st Rank in MATLAB Track)

2023

Developed a Domestic Waste Type Data analysis tool for a proposed Start-Up Solution

IIIT-Kalyani

National Level Basic Sciences Competitions:

ChemEnigma (1st Rank)

2025

Emerged champions at 72 hours Chemistry contest of theoretical, experimental and concept-presentation

IISc-Bangalore

All Bengal Chemistry Quiz (2nd Runners Up)

2025

Solved chemistry based questionnaires under time constraints

Presidency University

Mimansa (Zonal Topper)

2024

Contributed to the team in Mathematical problem solving

IISER-Pune

NAEST-National Anveshika Experimental Skill Test (Zonal Runners Up)

2023

Create extensive experimental setup using homely items- see [papers](#)

NANI, IIT Kanpur and Shiksha Sopan

Competitive Examinations:

Qualified JEE Mains and Advanced

2022

Ranked in Top 0.1 fraction of Candidates

Qualified IAT(IISER-Aptitude Test)

2022

Ranked in top 0.06 fraction of candidates

Qualified WBJEE
Ranked in Top 0.05 fraction of Candidates

2022

Scholarships and Honors:

Reliance Foundation Undergraduate Scholar
Qualified the RF-UG Aptitude test to be in the top 5000 students to be awarded this

2023

Best Young Scientist Speaker on Nanotechnology
Successfully presented at the prestigious conference

2019

World Science Conference, Jadavpur University

Professional Experience

Freelance Web Developer – ChemActiva Innovations Pvt. Ltd. 
Nanocellulose deeptech startup; DST-NIDHI PRAYAS and HDFC Parivartan grantee

2024 – 2025

- Designed and built the company's public website end to end as a freelance engagement, covering information architecture, responsive layout, product and technology pages, and deployment.
- Live at chemactiva.com.

Web Development for Anicon 3.0 
Inquivesta XI, IISER Kolkata

2024-2025

- Developed and led the web development of Anicon 3.0 Event.

Web Development for Material Science Laboratory 
EFAML, IISER Kolkata

2025

- Developed and designed the researchers lab info page available under Dr. Soumyajit Roy's homepage.

Private Educator & Technical Trainer
Advanced Computing & Basic Sciences Instruction (Self-Employed), Kolkata

2018-Present

- Developed and conducted courses in Computer Science, Physics, Chemistry and English for High School students (ICSE, CBSE, WB Board)
- Mentored 50+ students for Board and competitive examinations

Technical Consultant
Self-Employed, Kolkata

2021-Present

- Designed and implemented high-performance computing solutions
- Expertise in system optimization, BIOS/UEFI configuration, and OS installation
- Successfully completed 50+ custom build projects with 100% client satisfaction

Published Author
MindScapes (ISBN: 978-9389923209), Kolkata

2020

- Published creative anthology work focusing on metaphorical and philosophical themes
- Conducted workshops on technical and creative writing

Leadership & Community Impact

President, Slashdot - The Programming and Design Club 
IISER Kolkata – elected with a 99% approval vote of the club membership

Aug 2025 – Present

- Elected to lead the official programming and design club of IISER Kolkata with a **99% mandate**, one of the strongest endorsements recorded for the post, reflecting sustained contribution to the club before taking office.
- Directing the club's technical programme: workshops, hackathons, machine-learning and local-LLM sessions, and the club's open-source and web presence, building a durable coding culture on campus.

Office Bearer, Valence - The Chemistry Club
IISER Kolkata

Aug 2025 – Present

- Leading the official chemistry club of IISER Kolkata, organizing seminars, student talks, and departmental events to foster a vibrant chemical science community.

Organizer, Qiskit Fallfest 2025
Sponsored by IBM Quantum

Oct 2025

- Organized the local chapter of Qiskit Fallfest (one of 150 selected institutes globally sponsored by IBM); conducted hands-on workshops and tutorials on Qiskit SDK and quantum computing fundamentals.

Event Management & Logistics
IISER Kolkata

2023 – 2025

- Anicon 3.0 & 2.0:** Led organization of two editions of the sci-fi convention with 500+ participants.
- Academic Events:** Organized Supra-Molecular Discussions 2024 and GIAN Courses on Soft-Oxometalates and X-Ray Crystallography.

Social Impact & Community Service
Volunteering

2020 – 2021

- Served as a COVID-19 relief coordinator and currently mentoring students under the Ek-Pehal education initiative.

Additional Activities: District Level Debate and Quiz finalist | Shotokan Karate practitioner | 4th Year Art and Painting Student (Stroke Art & Portraits)

Honors & Awards

1st Prize (Rs. 2,00,000) – UIDAI Data Hackathon 2026

Jan–May 2026

Unique Identification Authority of India (UIDAI), NIC and MeitY, Government of India

- Won the **national Aadhaar data hackathon** outright: nearly **15,000 teams registered**, over **5,000 submitted solutions**, 30 projects were shortlisted and 15 finalist teams were assessed by the jury before the winning entry was announced at the valedictory session on **9 May 2026**.
- Winning team drawn from **IISER Kolkata** and the Institute of Engineering and Management, Kolkata. Delivered an in-depth analysis of UIDAI's aggregated Aadhaar enrolment and update datasets, characterising **biometric update patterns and enrolment trends across regions, states and demographic groups** and deriving predictive indicators.
- Produced concrete operational recommendations for improving Aadhaar service delivery; the CEO of UIDAI publicly noted that such work has “the potential to directly support policy and operational improvements”.
- **Coverage:** [Press Information Bureau](#) | [UIDAI](#) | [The Tribune](#)
- **Keywords:** large-scale data analysis, population-scale identity data, exploratory data analysis, geospatial and demographic segmentation, predictive indicators, public-policy analytics, data visualisation, Python, Pandas.

Invited Talks and Positions of Jury

Technical Talk – Running and Optimizing Local LLMs

Aug 2025

Slashdot Student Chapter, IISER Kolkata

- Delivered a hands-on session covering hardware setups, quantization methods, embedding pipelines, vector databases, and practical optimization techniques for local LLM deployment.

Invited Talk – Sustainability, AI, and Emerging Technologies

Aug 2025

Valence – Chemistry Society, IISER Kolkata

- Presented an interdisciplinary talk bridging sustainability science with AI-driven modelling, retrieval systems, and computational chemistry tools.

Track Co-organizer, FIRE 2026 – SYCO PHANCY Shared Task

2026

Explainable AI in Legal Reasoning: From Statute Prediction to Sycophancy Detection

- Co-organizing a shared task track at FIRE 2026 with international collaborators from IISER Kolkata, Université de Bretagne Occidentale (France), and University of Amsterdam (Netherlands), evaluating LLM interpretability and sycophancy detection in legal AI.

Guest Lecturer & Guest of Honour – CBSE District-Level Deliberation (DLD), Kolkata District

Aug 2026

CBSE STEM DLD workshop on Computational Thinking & AI, Abhinav Bharati High School (ABHS), Kolkata

- Invited by the Central Board of Secondary Education as an **AI subject expert** to address **64 teachers from 38 schools** across the Kolkata district on “AI in the Real-World Context” – what today’s AI tools and services actually are, the next-token-prediction machinery underneath them, and how to teach with and about them responsibly.
- Authored and released the full 60-minute talk openly in two formats: an interactive, pointer-driven [HTML slide deck](#) (26 slides, keyboard navigation, in-code SVG diagrams) and a [LaTeX Beamer PDF](#) with matching TikZ figures, under a public GitHub repository so attending schools can reuse the material.

Invited Judge – Voice of Youth 2026 National Summit & Hackathon

1–2 Aug 2026

Raajkutir, Kolkata; national teen-led innovation and leadership summit supported by Mr. Harshavardhan Neotia

- Served on the judging panel evaluating shortlisted student teams presenting at the National Offline Summit across the technology and innovation domains, assessing technical depth, feasibility, and demonstrated impact.

Domain Judge – Computer Science Events

July 2025

La Martiniere for Boys Annual Science Fest, Kolkata

- Invited to serve as an external judge for multiple Computer Science events involving project demonstrations, live coding, and problem-solving rounds.

Education

Indian Institute of Science Education and Research - Kolkata

CGPA: 8.2

BS-MS (Chemistry Major, Computer Science Minor)

2022-2027 (expected)

Calcutta University

CGPA: 8.308

B.Sc Honours in Physics (1st Year Only)

2021-2022

Jodhpur Park Boys' High School

83%

Higher Secondary Education in Physics, Mathematics, Chemistry, Computer Science (WBCHSE)

2019-2021

The New Horizon High School

83.75%

Secondary Level Schooling(English Medium) under WBBSE

2009-2019

References available upon request